

Epitope scaffolding with RFdiffusion3 + ProteinMPNN + AlphaFold3: pass rates and a native-antigen-aware accessibility surrogate

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Living manuscript — regenerated from `results/`; see `docs/WORKFLOW.md`

Abstract

We scaffold target epitopes onto de novo proteins and validate them with AlphaFold3. For antigens with a known antibody (the “DP3” set) we score designs against a four-filter ground truth; for antigens without an antibody (a tiled 12-mer set) we replace the antibody-clash term with a native-antigen-aware cylinder surrogate that measures whether scaffold mass intrudes into the space an antibody would need. A single config-driven composite scorer ranks designs in both settings. This document is the living record of the project: every number and figure is regenerated by a named script from `results/`.

1 Introduction

Designing a protein that displays a target epitope in its native conformation is a route to immunogens and binders that recapitulate a known antibody contact. We test the RFdiffusion3 → ProteinMPNN → AlphaFold3 pipeline against two questions.

Backbone geometry vs. sequence distribution (DP3 set). An RFdiffusion3-only pipeline passes AlphaFold3 validation far less often than Lawson’s RFdiffusion1 + ProteinMPNN baseline. The competing explanations are that RFdiffusion3 produces poorer *backbone geometry*, or that its *sequences* are out-of-distribution for AlphaFold3. Re-sequencing the RFdiffusion3 backbones with ProteinMPNN (trained on the inverse-folding task, sharing AlphaFold3’s natural-PDB distribution) isolates the two: if RFD3+MPNN approaches the baseline pass rate, the sequence-distribution hypothesis is supported. The four-filter definitions and run layout are recorded in `docs/README_v2_original.md`.

Scaffolding without an antibody (12-mer set). Many targets have no known antibody, so there is no antibody-clash filter. We instead scaffold tiled 12-residue epitopes and judge antibody *accessibility* with a native-antigen-aware cylinder surrogate (Section 2): it asks whether non-epitope scaffold residues intrude into the volume an approaching antibody would occupy, while excusing residues that merely sit where the native antigen already sits.

A single config-driven composite scorer (Section 2) ranks designs in both settings, differing only by which accessibility term it uses.

2 Methods

2.1 Pipeline

RFdiffusion3 generates scaffold backbones around the epitope; ProteinMPNN re-sequences them; AlphaFold3 predicts the structure single-sequence (`--norun_data_pipeline`, one diffusion sample, seed 1). Per-design metrics are computed from the AlphaFold3 prediction against the designed structure. Generation code is in `episcaf_pipeline/`; metrics/scoring/viz in `episcaf_analysis/`.

2.2 Four-filter ground truth (DP3 set)

A design passes when all four hold: overall backbone RMSD ≤ 2 Å; epitope-chunk RMSD ≤ 1 Å; mean predicted aligned error < 5 ; and zero antibody residues within 4 Å of non-epitope scaffold residues (`af3_n_clash_res`). Clash detection Kabsch-aligns the AlphaFold3 epitope onto the true epitope, then checks non-epitope scaffold heavy atoms against antibody atoms.

2.3 Native-antigen-aware cylinder surrogate (12-mer set)

With no antibody available, we approximate the occluded volume with a cylinder fitted to the epitope. The epitope C α atoms define a plane by SVD; the outward normal (the antibody approach direction) orients a cylinder of radius 16 Å, height 40 Å, offset -4 Å. We count non-epitope scaffold C α atoms inside the cylinder. The *native-aware* variant excludes scaffold atoms within a short distance of a native-antigen heavy atom — i.e. residues sitting where the native antigen already sits, which a real antibody tolerates. The geometry core is `episcaf_analysis/native_cylinder_core.py` (self-tested).

2.4 Composite scorer

One config-driven scorer (`episcaf_analysis/score.py`) ranks designs: (i) **gate** on epitope-chunk RMSD; (ii) transform each metric to a “higher-is-better” score within a scope (pooled, or per antigen); (iii) **weighted sum**; (iv) keep the **top- k per epitope**. Each metric is one input with its own activation (percentile / min-max / z-score / sigmoid), so the model is a single-layer perceptron with hand-set weights — “feed-forward, no backprop”. The dials live in `episcaf_analysis/presets.py`:

metric	weight	accessibility term
epitope-chunk RMSD	0.35	—
epitope PAE	0.25	—
overall RMSD	0.15	—
accessibility	0.25	<code>cylinder_native_aware</code> (12-mer) / <code>cylinder_ca_clashes</code> (DP3)

The 12-mer preset scores per antigen and keeps top-5 per epitope; the DP3 preset scores pooled and keeps top-15. Weights are a starting point; Section 4 describes fitting them to the DP3 `is_pass` label.

3 Results

All numbers below are the current findings and are regenerated from `results/`; the figure-generating commands are recorded in `figures/FIGURES.md`. Values still to be re-confirmed from a committed `results/` table are marked `[regen]`.

3.1 12-mer epitopes pass more often than DP3 epitopes

Using epitope-chunk RMSD < 2.5 Å as a common bar, the tiled 12-mer antigens all pass more often than the pooled DP3 / antibody set:

set	pass rate	median epitope RMSD (Å)
DP3 / mAb (pooled)	30.0%	5.0
6M0J	47.0%	2.65
1D2K	60.2%	1.85
4WAT	84.4%	0.43

Even the hardest 12-mer antigen (6M0J) clears the bar more often than DP3. Re-running the scorer’s gate over the full 12-mer set reproduces this: 54 539 of 82 712 designs pass ($\approx 65.9\%$ pooled), consistent with the per-antigen rates above. The full distribution comparison is in Fig. 1.

figures/dp3_vs_12mer_ungated.png
(not yet regenerated — see figures/FIGURES.md)

Figure 1: Ungated metric distributions, DP3 vs. the three 12-mer antigens, with per-row n and the “% < 2.5 ” pass rate annotated. [regen]

Caveats (important for interpretation). The 12-mer epitope is a single contiguous 12-residue stretch, whereas DP3 epitopes are larger and often multi-island, so 12-mer RMSD/PAE are mechanically lower in part by *definition*; the pooled DP3 30% also averages over heterogeneous epitopes. The clean control is to stratify *within* DP3 by island count/size (Section 4). Cylinder counts are also not comparable across sets: 12-mer cylinder medians (~ 31 – 32) exceed DP3 (~ 21) mainly because the longer scaffolds place more C α atoms in a fixed cylinder — a scaffold-size artifact, not more clashing. As a sanity check, the DP3 cylinder median (≈ 21) tracks its real `af3_n_clash_res` median (≈ 19).

3.2 The native-aware carve removes false-positive penalties

Among truly clash-free designs, excluding scaffold residues that sit in native-occupied space lowers the cylinder count and, in particular, cuts the heavily-penalized high-count tail, while barely changing genuinely clashing designs. The communication framing for the result: *~730 designs the antibody accepts were being penalized for residues sitting where the native antigen already sits — they no longer are.* See Fig. 2.

4 Open questions and next steps

1. **Within-DP3 island stratification (the clean control).** Split the DP3 set by epitope island count and size and recompute pass rates, to separate the “contiguous-12 is easier”

`figures/fp_reduction.png`
(not yet regenerated — see figures/FIGURES.md)

Figure 2: Distribution shift from the native-aware carve among clash-free designs. [regen via `episcaf_analysis/viz/plot_fp_reduction.py`]

confound from a real scaffolding-difficulty effect. Needs the per-epitope island metadata in `datasets/dp2.parquet`.

2. **Learn the scorer weights.** The DP3 set carries the four-filter `is_pass` label. Fit a logistic regression of the transformed metrics on `is_pass` to replace the hand-set weights, then transfer them to the 12-mer preset where no label exists.
3. **Confirm the canonical DP3 metrics builder.** Determine whether `compute_metrics_mpnn.py` (currently archived) is the live RFD3+MPNN metrics producer and promote it into `episcaf_analysis/` if so (see `docs/MIGRATION.md`).
4. **Validate the DP3 scoring path** end-to-end with `--preset antibody`, confirming `cylinder_ca_clashes` is the correct accessibility column there.
5. **Finish the 12-mer AlphaFold3 data migration** and verify landing counts against the manifest.

References